FRC PROGRAMMING CURRICULUM — MODULE 1, LESSON 01

Setup + First Drive

Get every student installed, building, and driving a simulated robot before they ever touch hardware.

Instructor Edition — Not for student distribution

At a Glance

Field	Detail
Target audience	HS freshmen & sophomores, brand new to FRC programming
Hardware	Student laptops only — no XRP needed today
Session length	2–3 hours (one extended session, not 45 minutes)
Key tools	WPILib installer (includes AdvantageScope), USB drive with starter project

Learning Objectives

- Students have WPILib + VS Code installed and can launch WPILib VS Code
- Students have AdvantageScope installed (included with WPILib) and can open it
- Students can build the First Drive project to BUILD SUCCESSFUL
- Students can launch the simulator, enable teleop, and drive with the keyboard
- Students can connect AdvantageScope and watch logged values change in real time
- Students can view the robot pose in AdvantageScope's 2D/3D view
- Students know where the reference card lives and have skimmed it once

Before You Start

USB drive contents — prepare ONE per student (or one per pair)

- WPILib installer — Windows .iso + VS Code .zip (offline install)
- Starter project ZIP — Lesson01-Project.zip
- Reference card PDF — XRP-Reference-Card.pdf

Download locations

- WPILib: https://docs.wpilib.org/ → Installation
- AdvantageScope is bundled with WPILib — no separate download needed

Backup laptops

- Have 2–3 pre-configured laptops ready with everything installed
- Use these for students whose installs fail — troubleshoot personal laptops later

Network

- If possible, disable school firewall/content filtering for this session
- If not, USB-drive install works — Gradle dependencies will still need internet on first build

The most important instructor mindset for this lesson

Setup failures are not a moral failing. They are the cost of using real software on real laptops in a real school. Resist the urge to make students troubleshoot their personal machine for an hour. Switch to a backup laptop after 15 minutes of stuck. The goal today is everyone leaves with a working setup — not necessarily on their own laptop.

Session Timing

Time	Phase	You do	Students do
0–10 min	Hook + setup	Frame the day. Hand out USB drives. Verify everyone has one.	Plug in laptop. Copy USB contents to Documents folder.
10–60 min	Install WPILib	Live-demo on projector. Circulate. Triage stuck students into backup laptops after 15 min.	Run installer. Choose options. Wait. Launch WPILib VS Code 2026. Verify AdvantageScope opens (bundled).
60–100 min	Build the starter	Demo: open folder, build, see BUILD SUCCESSFUL. Watch for VS-Code-vs-WPILib-VS-Code mistakes.	Extract ZIP. Open in WPILib VS Code. Build. See BUILD SUCCESSFUL.
100–150 min	First drive	Demo simulator + AdvantageScope flow on projector first. This is the payoff — everyone must reach it.	Launch sim. Enable teleop. Drive with W/S/A/D. Launch AdvantageScope via Start Tool. Connect. Graph velocity values.
150–170 min	2D/3D pose view	Demo adding robot pose to AdvantageScope's 2D/3D tab. Include arm and XRP model setup.	Add pose field + arm component. Set custom assets. Select XRP robot model. Watch robot move on screen.
170–180 min	Reference card + wrap	Walk through reference card sections. Run end-of-session checklist student by student.	Save reference card. Run through checklist. Pack up laptop.

90-minute compressed version

If you only have 90 minutes (single class period), accept that some students will leave with incomplete setups. Skip the 2D/3D pose view and the reference card walkthrough. Get everyone through Install → Build → First Drive at minimum. Schedule a follow-up for the pose view and the reference card before Lesson 02.

Phase 1 — Hook + Setup (0–10 min)

Set expectations and hand out materials.

Opening

Don't pretend this isn't a slow session. It is. Tell students up front.

Script — what to say

"Today's goal is simple: get everything installed and see your first robot code run. Installations fail sometimes — that's normal. If yours doesn't work, use one of the backup laptops and we'll troubleshoot your own laptop later. The important thing is everyone leaves today with a working setup."

Hand out USB drives

- Verify every student has a USB drive (or knows which pair-partner has one)
- Tell students to copy the contents to their Documents folder before running anything
- Don't run installers from the USB drive — copy first, then run

Phase 2 — Install WPILib (10–60 min)

The slow one. Plan for 50 minutes. Students will have time to chat — that's fine. Your job is to triage failures.

What students do

- Run the WPILib installer (.iso on Windows — mount it, then run the .exe inside)
- Select the existing VS Code archive for offline install when prompted
- Accept Windows SmartScreen warnings
- Choose options: keep default install path, install everything
- Wait 10–15 minutes for extraction
- Launch WPILib VS Code 2026 (purple theme = correct one)
- Open AdvantageScope (bundled with WPILib — no separate install needed)

What you do

- Circulate constantly. Don't sit at the front.
- Watch for the common issues below. Most show up early.
- After 15 minutes of stuck on one issue, redirect the student to a backup laptop

Common issues — WPILib

Issue	Solution
"Not enough disk space"	Free up ~5 GB or use a backup laptop.
Installer hangs at extraction	Wait longer — first install is genuinely slow. After 20 minutes of no progress, restart it.
VS Code won't launch after install	Restart computer, then try again. If still failing, use a backup laptop.
Windows SmartScreen blocks installer	Click "More info" → "Run anyway".
Already had VS Code; opened wrong one	WPILib VS Code is purple-themed. Pin it to taskbar to avoid the mistake again.
AdvantageScope won't open	It's under the WPILib install folder. Check Start menu for "AdvantageScope".

Pacing check — by 60 minutes — Most students should have WPILib installed, WPILib VS Code launched, and AdvantageScope verified open. Anyone still stuck should be on a backup laptop now.

Phase 3 — Build the Starter Project (60–100 min)

First time real code touches their laptop. Watch for the WPILib-VS-Code-vs-regular-VS-Code mistake. Demo on projector first.

Demo on the projector first

- Show extracting the ZIP into Documents folder
- Show File → Open Folder in WPILib VS Code, navigate to Lesson01-Project\Lesson01\First Drive
- Show "Yes, I trust the authors"
- Show Ctrl+Shift+P → WPILib: Build Robot Code
- Show terminal output ending in BUILD SUCCESSFUL

Then students replicate

Walk the room. Catch these mistakes early:

- Working from inside the ZIP without extracting
- Opening a single file instead of the project folder
- Using regular VS Code (no purple theme) instead of WPILib VS Code
- Cancelling the Gradle dependency download because "it's slow"

Common issues — Starter project

Issue	Solution
"Could not find build.gradle"	Student opened a single file, not the folder. Use File → Open Folder.
Build fails with Gradle errors	Check internet — first build downloads dependencies (~2–5 min).
"Java not found"	Opened regular VS Code, not WPILib VS Code. Switch.
Build runs but no terminal output	Terminal panel collapsed. Click the Terminal tab at bottom.
"Trust the authors" dialog never closes	Click in the editor area first to give it focus, then click Yes.

Pacing check — by 100 minutes — Everyone has BUILD SUCCESSFUL on their screen. If anyone is still stuck on the build, switch them to a backup laptop now — they need to make it to the first drive in this session.

Phase 4 — First Drive + AdvantageScope (100–150 min)

This is the payoff. Make sure everyone gets here. Demo every step on the projector first — then watch students replicate. The wow moment is when the AdvantageScope graph spikes as they hold W.

Live demo sequence

Run this top-to-bottom on the projector. Don't skip steps. Students follow along on their own laptops afterward.

Step 1 — Launch the simulator

- Ctrl+Shift+P → WPILib: Simulate Robot Code
- Check halsim_gui → OK
- Wait for build (fast — already compiled)
- Simulation GUI window opens

Step 2 — Set up keyboard driving

- In Simulation GUI → System Joysticks panel → drag Keyboard 0 to Joystick[0] slot
- Robot State panel → Teleoperated → Enable
- Click Simulation GUI window — keyboard only works when this window has focus
- Keys: W forward, S backward, A turn left, D turn right

Step 3 — Open and connect AdvantageScope

- In WPILib VS Code: Ctrl+Shift+P → type WPILib: Start Tool → select AdvantageScope
- In AdvantageScope: File → Connect to Simulator → select Default: NetworkTables 4 (or Ctrl+Shift+K)
- Confirm connection: the time bar at the top of AdvantageScope starts running
- The left sidebar populates with AdvantageKit/Drive/, AdvantageKit/DriverStation/, etc.
- Connect BEFORE driving — otherwise the first seconds of data are lost

Step 4 — Graph velocity values

- In AdvantageScope: select the Line Graph tab at the top
- In the left sidebar, expand AdvantageKit → Drive
- Drag LeftVelocityMetersPerSec onto the Left Axis below the graph area
- Drag RightVelocityMetersPerSec onto the same Left Axis
- Switch to Simulation GUI, hold W for 2–3 seconds, release
- Switch back to AdvantageScope — both lines spike up, then drop to zero
- Ask: 'What happens if you press A or D?' Let them try.

Common issues — First drive

Issue	Solution
Simulation GUI doesn't appear	Check for popup blocker. Window may have opened off-screen — drag it back.
Robot doesn't respond to keys	Two causes: (1) Teleop not enabled, (2) Simulation GUI doesn't have focus. Click it.
AdvantageScope won't connect	Simulator must be running first. Restart sim, then connect.
Connected, but no values appear	Robot must be enabled in teleop. Disabled robots don't publish data.
Values exist but graph stays flat	Wrong fields dragged. Use AdvantageKit/Drive/LeftVelocityMetersPerSec, not Setpoints. Also confirm Line Graph tab is selected.

Pacing check — by 150 minutes — Every student has seen velocity values spike on the AdvantageScope graph. If anyone hasn't, stop the lesson and get them there before moving on.

Phase 5 — 2D/3D Pose View (150–170 min)

The bonus wow. AdvantageScope can render the robot's position on a field view. This is where simulation stops feeling abstract.

What students do — 2D Field

- In AdvantageScope: select 2D Field from the top tabs
- In the left sidebar, expand AdvantageKit → RealOutputs → Drive
- Find Robot (shows as Pose - Pose2d) → drag it onto the 2D Field view under Poses
- A blue box with an arrow appears in the top-right corner of the field — this is the Blue Origin
- Click the Simulation GUI and drive with W/S/A/D — the robot icon moves on the field

What students do — 3D Field

- In AdvantageScope: select 3D Field from the top tabs
- The pose does not carry over automatically — expand AdvantageKit → RealOutputs → Drive → drag Robot under Poses
- Also drag FinalComponentPoses (shows as Pose3d) into the same Poses box — this adds the arm
- Go to App → Use Custom Assets Folder and select the custom assets folder provided (contains the XRP robot model)
- Close and restart AdvantageScope to load the custom assets, then reconnect to the simulator
- In the 3D Field view, right-click the robot pose entry and select XRP Robot
- Click the Simulation GUI and drive with W/S/A/D — control the arm with Z / X
- 3D mouse controls: left-click drag = rotate, scroll wheel = zoom, right-click drag = pan
- AdvantageScope saves the pose configuration — next session it will already be set up

Common issues — 2D/3D pose

Issue	Solution
No Pose field visible	Expand AdvantageKit → RealOutputs → Drive. Robot must be enabled.
Robot doesn't move on field	Verify Robot (Pose2d) is dragged — not a velocity or encoder field.
2D view shows robot but it's off the field	Field image may need to be selected. Check AdvantageScope's field dropdown.
3D view not responding to mouse	Click inside the 3D view first to give it focus.
XRP Robot not in model list	Custom assets folder not loaded. Restart AdvantageScope after setting the folder.
Arm doesn't move in 3D	FinalComponentPoses not added to the Poses box. Add it alongside the drive Robot pose.

Pacing check — by 170 minutes — Students have seen the robot and arm move on the 2D/3D view. If time is short, this phase is the first to cut.

Phase 6 — Reference Card + Wrap-Up (170–180 min)

Cool down. Students are tired. Make this section short and concrete — checklist, save the reference card, send them home with a clear sense of what they accomplished.

Reference card walkthrough

Open the reference card on the projector. Spend 5 minutes pointing at sections — not reading them. The goal is recognition, not memorization.

- XRP Drivetrain methods — "this is where you'll look up arcadeDrive() in Lesson 04"
- Unit conversions — "this prevents the most common bug in Lesson 04"
- AdvantageKit logging pattern — "the three lines you'll write in every subsystem"
- CommandScheduler — "the line that must be in robotPeriodic()"

End-of-session checklist

Walk through each student. Verify each item before they pack up:

- ☐ WPILib VS Code opens (purple theme)
- ☐ AdvantageScope opens
- ☐ First Drive project builds successfully
- ☐ Simulator runs and responds to keyboard
- ☐ AdvantageScope connects (time bar starts) and shows velocity values when driving
- ☐ AdvantageScope 2D field view shows robot moving
- ☐ AdvantageScope 3D field view shows robot and arm moving with XRP model
- ☐ Reference card saved

Script — what to say

"You just installed a complete development environment, compiled real robot code, drove a simulated robot, watched live data update on a graph, and saw the robot move on a 2D field view. That's a lot. Next lesson, we'll look at what VS Code actually created for us — and why the code is structured the way it is. Bring your laptop, your power cord, and the reference card."

Common Student Questions

Q: Why do we need both VS Code and AdvantageScope?

A: "VS Code is where you write code. AdvantageScope is where you see what the robot is doing. They're different tools for different jobs. You'll have both open most of the time."

Q: When do we use the real XRP robot?

A: "Lesson 04. Today is about getting your laptop ready. The simulator lets us write and test code before we touch any hardware — that's how real teams work too."

Q: What if I already have VS Code installed?

A: "WPILib VS Code is different from regular VS Code — it has robot-specific tools built in. You need the WPILib version. Pin it to your taskbar so you don't accidentally open the wrong one."

Q: My install is stuck. Should I cancel?

A: "If the progress bar moved at all in the last 10 minutes, no — wait. If it's been frozen for more than 15 minutes with no disk activity, cancel and try again. After two failed attempts, switch to a backup laptop."

Q: Do I have to bring my laptop every session?

A: "Yes. Every session. With the power cord. The whole curriculum is hands-on — there's no useful version of these lessons without your laptop."

Post-Session — Your Homework as Instructor

Send to students

- A reminder to bring laptop + power cord + reference card to Lesson 02
- Backup laptop assignments — who's borrowing which machine
- Optional: link to the digital handout (lesson-01-setup.html) for review

Track for next session

- Which students are on backup laptops
- Which students had major issues — schedule 1-on-1 troubleshooting before Lesson 02
- Lessons learned — record any issues that should be fixed before next year's Lesson 01

Lesson 01 Success Metrics

By the end of this session:

- 80%+ of students have installations working on their own laptops
- 100% of students have a working setup — own laptop or backup
- 100% of students have seen the simulator run successfully
- 100% of students have seen velocity values graph in AdvantageScope
- 100% of students have seen the robot move in AdvantageScope's 2D field view
- 100% of students have seen the robot and arm move in AdvantageScope's 3D field view with the XRP model

> If these metrics aren't met, do not proceed to Lesson 02 yet. Schedule a follow-up session and finish the setup. Lesson 02 assumes a working environment.